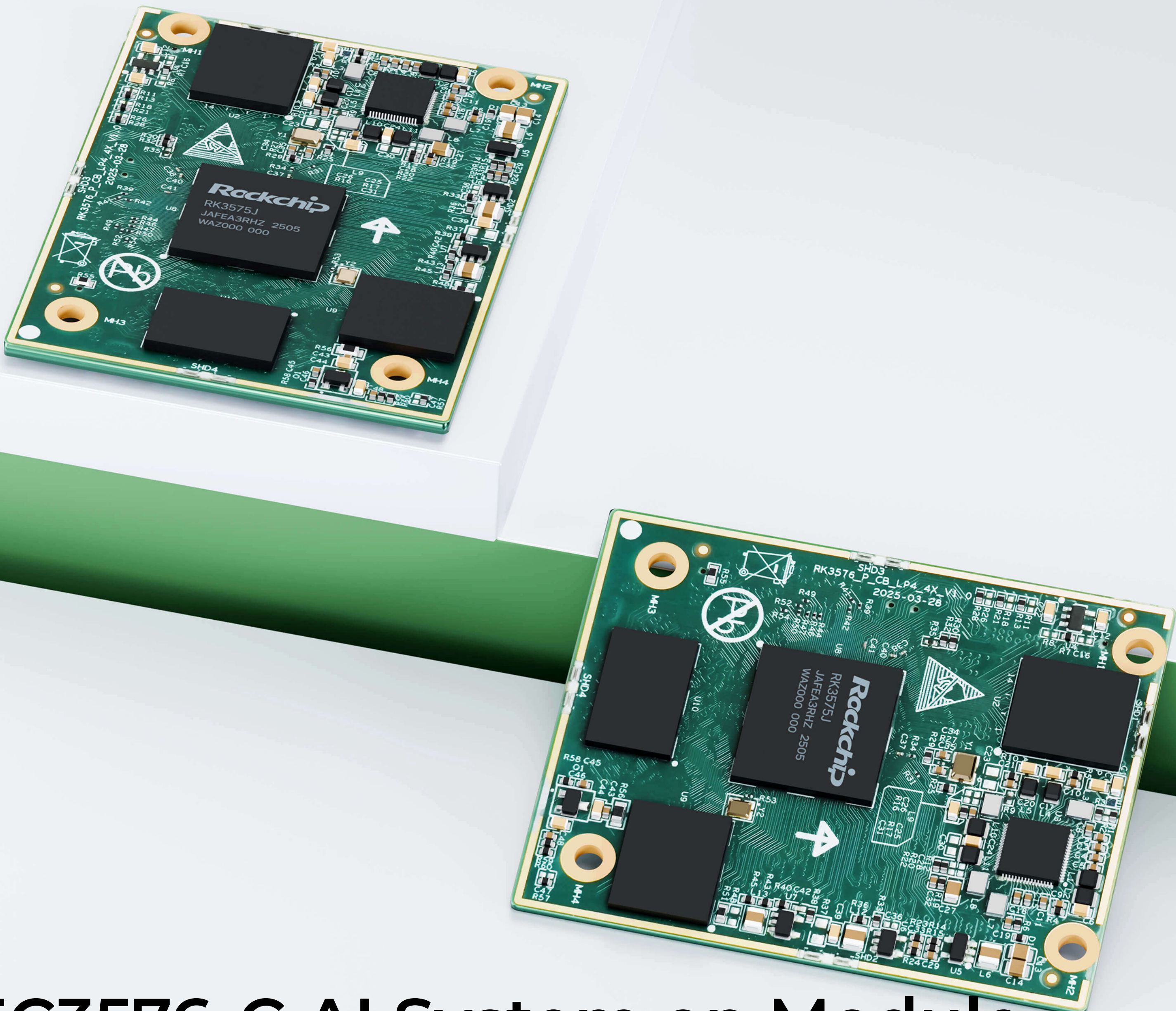
InHand Networks

Embrace Edge AI, Empower Industrial Digitalization



EC3576-C AI System on Module

- RK3576J ARM SoC
- 4×A72+4×A53
- 6TOPS NPU
- 8K Video Codec

1. Product Overview

EC3576-C system on module is powered by the Rockchip RK3576 processor, tailored for AIoT and industrial markets.

Product Features:

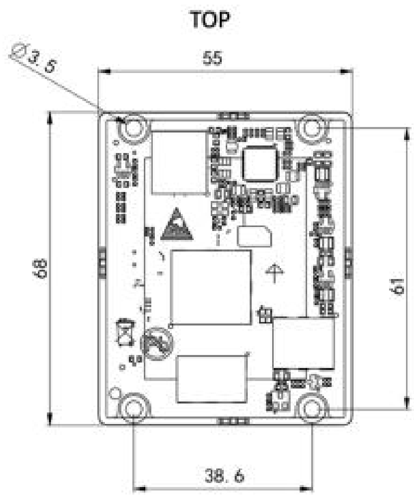
- **High-Performance CPU:** 8nm octa-core, 4×A72+4×A53 @ 2.3 GHz
- **Powerful AI NPU:** 6.0 TOPS INT8, multi-format support
- **Advanced Graphics:** Mali-G52 MC3, 8K codec, triple display
- **Rich Interfaces:** PCIe, USB 3.2, CAN-FD, GMAC, MIPI
- **Industrial Grade:** -40 °C ~ +85 °C, ultra-thin connector

Core Technical Specifications

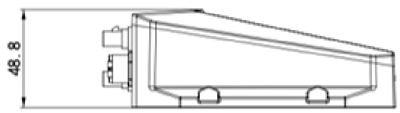
Technical Indicator	Specification
OS	Android 14
Industrial Protocol	Modbus RTU/TCP, EtherNet/IP, OPC UA, Mitsubishi MC
Remote Management	InHand DeviceLive, HTTPS, SSH
AI	6.0 TOPS INT8 NPU
Video Codec	8K@30 fps decode, 4K@60 fps encode
Cloud Platform	Cloud parameter config, container management, firmware management
CPU	4×A72 @ 2.3 GHz + 4×A53 @ 2.2 GHz
Memory / Storage	8 GB LPDDR4 / 256 GB UFS
Display Interface	HDMI/eDP/DSI/DP, triple independent display
Connectivity	2×GMAC, USB 3.2, CAN-FD, MIPI-CSI×5, UART×12

Technical Indicator	Specification
Dimensions (W × D × H)	55 × 68 × 3 mm
Operating Temperature	-40 °C ~ +85 °C

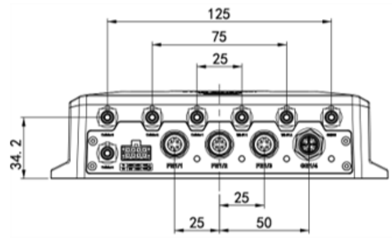
2. Product Dimensions



Front View



Side View



Bottom View

Note:
1. All dimensions are in millimeters (mm).
2. All dimensions are approximate and for reference only.
3. Dimensioned drawings are not intended for machining.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

3. Hardware Specifications

Category/Parameter	Specification
Hardware Platform	

Category/Parameter	Specification
CPU	4 × ARM Cortex-A72 @ 2.3 GHz + 4 × ARM Cortex-A53 @ 2.2 GHz
NPU	6.0 TOPS INT8, supports INT4/INT8/INT16/FP16/BF16/TF32
GPU	ARM Mali-G52 MC3
VPU	Hard Encode: H.264, H.265, 4K@60 fps; Hard Decode: H.264, H.265, VP9, AV1, AVS2, 8K@30 fps or 4K@120 fps
RAM	8 GB LPDDR4
ROM	256 GB UFS
Display Interface	
HDMI/eDP TX	1 × HDMI v2.1 / eDP v1.3 (muxed), up to 4K@120 Hz; Supports CEC, ARC, HDCP v2.3/v1.4; eDP supports up to 3 displays with different content
MIPI DSI	1 × MIPI DSI-2 TX, D-PHY v2.0 (4 lanes) or C-PHY v1.1 (3 trios), up to 2560 × 1600@60 Hz
Parallel	1 × Parallel RGB/BT.656/BT.1120, up to 1920 × 1080@60 Hz
EBC	1 × E-ink EPD, 2560 × 1920 hard decoding, 16-bit data bus, up to 32-level grayscale
DP TX	1 × USB/DP combo, DisplayPort v1.4, up to 4K@120 Hz, supports MST, USB Type-C with DP Alt, HDCP v2.3/v1.3
Camera Interface	
MIPI-CSI	5 × CSI-2; 4 × 2-lane D-PHY v1.2 (2.5 Gbps/lane, combinable to 2 × 4-lane); 1 × 4-lane D-PHY v2.0 (4.5 Gbps/lane) or 3 C-PHY trios; Supports up to 5 cameras working together
DVP	1 × 8/10/12/16-bit standard DVP, up to 150 MHz data input; Supports BT.601/BT.656/BT.1120

Category/Parameter	Specification
Audio	
SAI	≤5; SAI 0/1 support 4 TX lanes + 4 RX lanes; SAI 2/3/4 support 1 TX lane + 1 RX lane; Supports I2S/TDM/PCM mode, sampling rate up to 192 kHz, 16~32 bits
Digital Audio Codec	1; Supports 2 × DAC, 3 blending modes each; Supports I2S/PCM master/slave mode, 16-bit sampling rate; Supports volume control
Connectivity	
Ethernet	≤2 × GMAC, pinout by RGMII/RMII, 10/100/1000 Mbps
SDIO	≤2, SDIO v3.0, 4-bit
USB 3.2	2 (1 × Type-C supports DP Alt mode; 1 × combo high speed interface)
USB 2.0 OTG	2
UART	≤12, 64-bit FIFO for TX/RX; Supports 5/6/7/8-bit serial transceiving, up to 4 Mbps; All 12 UARTs support flow control and RS485
CAN-FD	≤2, compliant with CAN and CAN-FD; Supports standard and extended frames; 8192-bit FIFO
SPI	≤5, master and slave mode, each supports two chip selections
I2C	≤9, 7-bit and 10-bit address modes; Standard mode 100 kbps, HS mode 400 kbps
PWM	≤16, supports interrupt operation, capturing mode
Power	
Input Power	DC 5V

Category/Parameter	Specification
Mechanical	
Dimensions (W × D × H)	55 × 68 × 3 mm
Package	Board-to-board connector (4 × 100-pin, 0.4 mm pitch, combined height 1.5 mm)
Mounting Holes	4 × ϕ3.5 mm
Environment	
Operating Temperature	-40 °C ~ +85 °C

4. Software Specifications

Category/Parameter	Specification
Operating System	
OS	Android 14
OS Flashing Method	USB OTG
Data Acquisition Protocol (DSA)	
Industrial Protocol	Modbus RTU Master/Slave, Modbus TCP Master/Slave, EtherNet/IP, ISO on TCP, OPC UA Client/Server, Mitsubishi MC 3C/3E/3C OverTCP, Mitsubishi CPU Port, FINS UDP, Host Link, PPI
Electricity Protocol	DLT645-2007, IEC101/104, DNP3.0
Other Protocol	BACnet, CNC

Category/Parameter	Specification
Maintenance and Management	
Upgrade Method	Supports patent upgrade mechanism, local or remote firmware upgrade
Log	Supports local system logs, remote logs, important log power-off preservation
Remote Management	InHand DeviceLive, HTTP, HTTPS, SSH, etc.
DeviceLive Cloud	Supports cloud-based parameter configuration, container management, application and firmware management

5. Ordering Information

Model Code

Model code: EC3576-C

Product Model

Model	CPU	CPU Clock Speed	RAM	ROM	Operating Temperature
EC3576-C	4 × A72 + 4 × A53	A7 @ 2.3 GHz, A53 @ 2.2 GHz	8 GB	256 GB	-40 °C ~ +85 °C

6. Contact Us

- **Website:** [InHand Networks]